

Suggested problems: 1, 4, 6-8, 10, 12, 17

Problem 1: Suppose that a committee composed of eight people is to be selected from a group of 20 people.

(a) How many different groups of people are there that might be on the committee?

(b) If each of the eight people on the committee is to be assigned one of eight different jobs, how many ways are there to choose the committee of eight people and assign them jobs?

Problem 2: Suppose that a committee of 12 people is selected in a random manner from a group of 100 people. Determine the probability that two particular people will both be selected.

Problem 3: Suppose that a fair coin is tossed 10 times.

(a) What is the probability of obtaining exactly three heads?

(b) What is the probability of obtaining three or fewer heads?